

# MU4MA329

---

## Approximation of elliptic PDEs and numerical simulation Programming sheet 3: Matrix assembly

**Question 1** Write a function `ElementaryMassMatrix` able to compute the elementary mass matrix for  $\mathbb{P}_1$  Lagrange finite elements for a single triangle.

**Question 2** Write a function `MassMatrix` able to compute the mass matrix for  $\mathbb{P}_1$  Lagrange finite elements on a triangular mesh. The format of the matrix should be a sparse format (using `scipy.sparse`). The numbering of the global degrees of freedom should use the one given from the coordinates array `vtx`.

**Question 3** Adapt the code of the elementary mass matrix to the case of a single edge (part of a generic 2D triangular mesh) and the code of the mass matrix to the case of the boundary of a triangular mesh. Use again the global numbering given by the coordinates array `vtx`, so that the size of the global mass matrix is larger than its rank.

**Question 4** Validate your code by numerically computing the areas and perimeters of simple shapes given in the mesh examples.

**Question 5** Write a function `ElementaryStiffnessMatrix` able to compute the elementary stiffness matrix for  $\mathbb{P}_1$  Lagrange finite elements for a single triangle.

**Question 6** Write a function `StiffnessMatrix` able to compute the stiffness matrix for  $\mathbb{P}_1$  Lagrange finite elements on a triangular mesh.

**Question 7** Validate your code by numerically computing  $\int_{\Omega} \nabla u \cdot \nabla v$  for two affine functions  $u$  and  $v$ .

**Question 8** Compute analytically and numerically the  $L^2$  and  $H^1$ -norms in the unit square  $\Omega = (0, 1)^2$  of the  $\mathbb{P}_1$ -interpolation of the function

$$u(x, y) := \sin(p\pi x) \sin(q\pi y), \quad \forall x, y \in (0, 1)^2, \quad (1)$$

for some  $p, q \in \mathbb{N}$ . What is the rate of convergence as you refine the mesh ?

**Question 9** Compute the run time of your assembly routines for increasingly large meshes. How do your timings scale with the number of degrees of freedom ?